

Assignment 4

Information Theory for Data Science

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Exercise 4.1 :

Stream Ciphers

Question 1 :

The 127 bit M-sequence is generated in the 'ex4_1.m' file, count of the ones and zeros are shown below.

As we can see $N_1 = N_0 + 1$

$N_0 =$
63
$N_1 =$
64

Question 2 :

For m cells : $\begin{array}{ccccccc} & & & & & & 1 \\ - & - & - & - & - & \dots & - \\ & & & & & & 0 \end{array}$

If the last bit (output) is $\downarrow$ equal to 1 ,
we have 2^{m-1} possibilities for the remaining
bits. Also, if the last bit equals to 0 ,
we have $2^{m-1} - 1$ possibilities.
all zero state $\leftarrow$

$$\Rightarrow \left. \begin{array}{l} N_1 = 2^{m-1} \\ N_0 = 2^{m-1} - 1 \end{array} \right\} \rightarrow N_1 = N_0 + 1$$

Question 3 :

The code for acquiring the table is in the 'ex4_1.m' file, and shown in the question 3 section. Output of the code is shown below:

NR0 =						
16	8	4	2	1	1	0
NR1 =						
16	8	4	2	1	0	1

Question 4 :

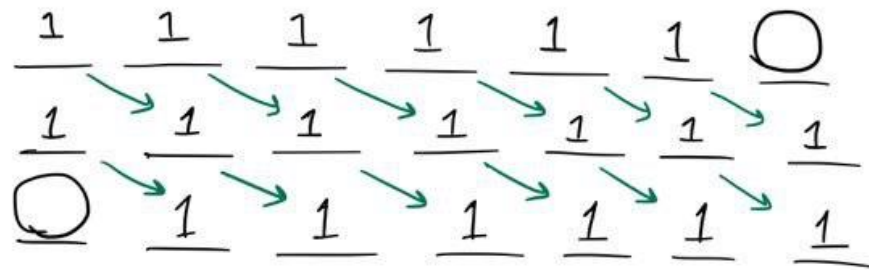
As we can see values of NR0 and NR1 satisfy $N_i = N_T/2^i$, which is the property of an ideal binary random sequence.

Also $N_T = N/2$.

Question 5 :

If we have a run of m zeros, we get stuck in this state and it can never change.

On the other hand, we can not have a run of $m+1$ ones.
Suppose this situation:



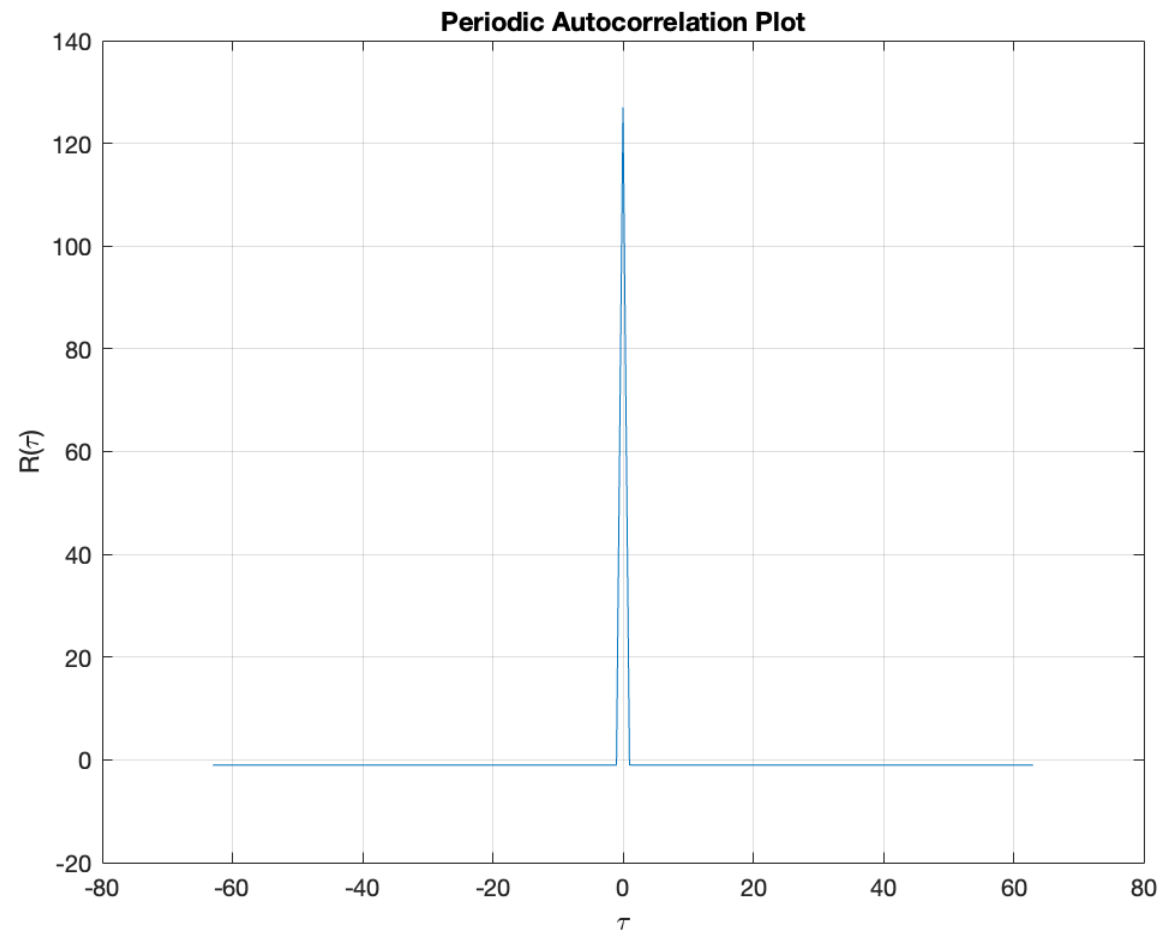
If we have a run of $m+1$ ones, we will have repetitive all ones states, so the cells shown with circle should be zero, and thus we can not have a run with $m+1$ ones.

Question 6 :

The code for acquiring the table is in the '*ex4_1.m*' file, and shown in the question 6 section. Output of the code is shown below, and we can see that

$$R(\tau) = N \text{ for } \tau = 0$$

$$R(\tau) = -1 \text{ for } \tau \neq 0 :$$



Question 7 :

The code for computing the primitive polynomial of the LFSR is in the '*ex4_2.m*' file, and shown in the question 7 section.

$X = 000000000000001001110$

$P(D) = D^{20} + D^6 + D^3 + D^2 + D \quad \leftarrow$ Primitive polynomial of the LFSR

Question 8 :

The code for generating the next 20 bits is in the '*ex4_2.m*' file, and shown in the question 8 section.

Next 20 bits is as below:

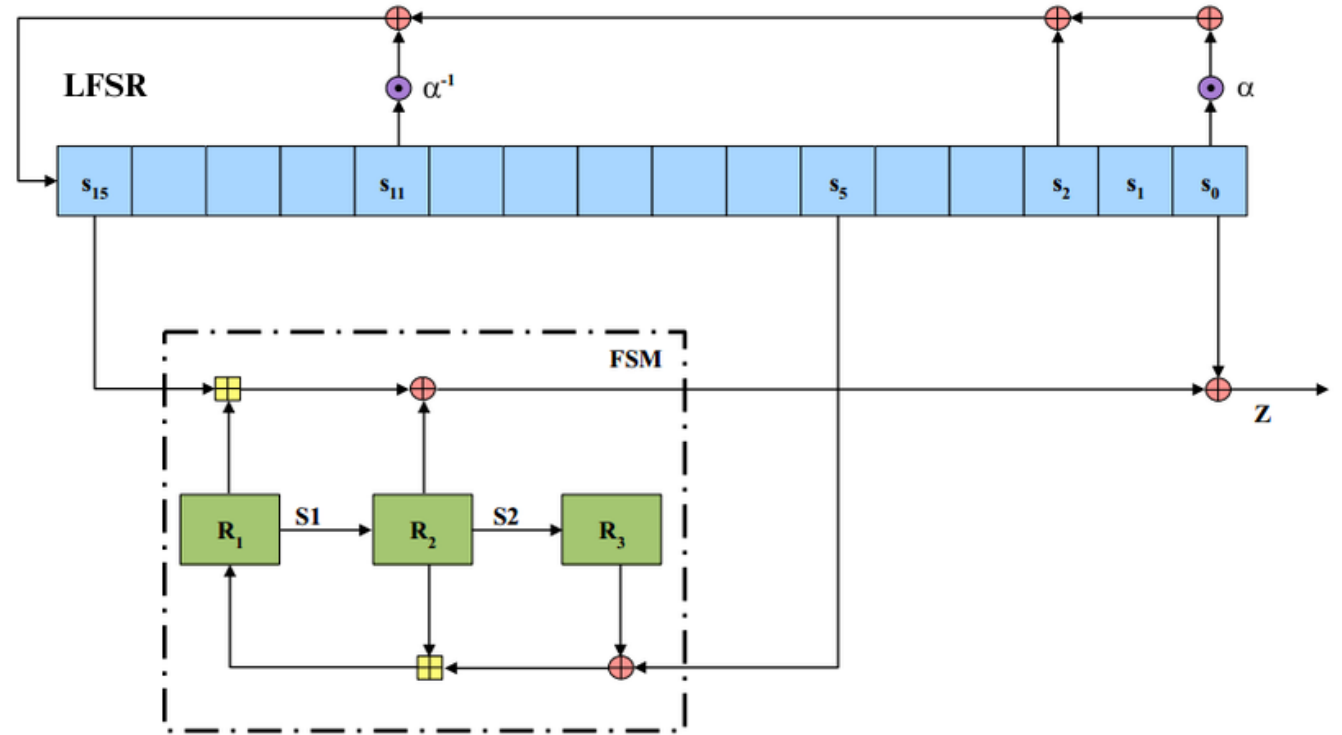
0 0 0 1 0 0 0 0 1 1 1 0 0 0 0 1 0 0 0 0

Exercise 4.1 Question 9 :

SNOW 3G stream ciphers

1. Introduction

SNOW 3G is a word-oriented synchronous stream cipher. It consists of a linear feedback shift register (LFSR) with 16 registers, each 32 bits in size, and a finite state machine (FSM) serving as a nonlinear component with three 32-bit registers.



2. Description

- SNOW 3G uses the LFSR to generate a pseudorandom key stream and then FSM is used as a filter function to introduce non-linearity to the cipher, enhancing its resistance to cryptanalysis.
- This key stream is XORed with the plaintext to generate the ciphertext.
- SNOW 3G incorporates periodic key updates to enhance security and prevent vulnerabilities associated with prolonged key usage.

3. Applications

- SNOW-3G is a stream cipher used by the 3rd Generation Partnership Project standards as the core part of the confidentiality and integrity algorithms for Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE) networks.
- The UMTS, LTE, and LTE-advanced standards are the 3rd- and 4th-generation mobile networks that enable mobile migrations of Internet applications like Voice over IP (VoIP), video streaming, music downloading, mobile TV, and so on. The security of these mobile standards is based principally on the SNOW-3G stream cipher.
- SNOW 3G is chosen as the stream cipher for the 3GPP encryption algorithms UEA2 and UIA2.

4. References

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Exercise 4.2.b :

Signal / TLS

1. The Signal Protocol

1. What is it?

The Signal protocol is a cryptographic messaging protocol that provides end-to-end encryption for instant messaging in WhatsApp, Wire, and Facebook Messenger among many others, serving well over 1 billion active users. Signal includes several uncommon security properties (such as “future secrecy” or “post-compromise security”), enabled by a novel technique called ratcheting in which session keys are updated with every message sent. [1] [3] [11]

2. How the keys are generated and exchanged:

- In WhatsApp all keys pairs are generated using *Elliptic Curves* using the *curve 25519*. With most encryption systems, when The app is installed on a phone, it creates a permanent key pair that is used to encrypt and decrypt messages: one "public" key that is sent to the messaging server and will be used to identify the user, and one "private" key that never leaves the user's phone. If that private key is somehow compromised, however, like if someone hacks or seizes your phone, that potentially leaves all your messages vulnerable to decryption. Even if you've deleted messages from your phone, the key can decrypt any encrypted messages that eavesdroppers have managed to record when they originally traveled across the network. [2] The Signal protocol, however, uses a so-called "ratchet" system that changes the key after every message. It does this by generating a collection of temporary key pairs for each user, in addition to the permanent keys. When someone sends a message to a contact over the app using the Signal protocol, the app combines the temporary and permanent pairs of public and private keys for both users to create a shared secret key that's used to encrypt and decrypt that message. Since generating this secret key requires access to the users' private keys, it exists only on their two devices. And the Signal protocol's system of temporary keys—which it constantly replenishes for each user—allows it to generate a new shared key after every message. [12]
- Signal uses the *Diffie-Hellman* key exchange protocol, specifically the *Ephemeral Diffie-Hellman (EDH)* variant. In this process, both users generate temporary, ephemeral key pairs. These temporary keys are used to derive a shared secret, known only to the communicating parties.

3. Which algorithms are used for encryption (no detailed description of the algorithm, but write on which technique they are based on (e.g., prime factorization or discrete log or elliptic curves or ...))

Elliptic Curve Cryptography (ECC):

It is an approach to public-key cryptography based on the algebraic structure of elliptic curves over finite fields. *ECC* allows smaller keys compared to *non-EC* cryptography (based on plain *Galois* fields) to provide equivalent security.

Elliptic curves are applicable for key agreement, digital signatures, pseudo-random generators and other tasks. Indirectly, they can be used for encryption by combining the key agreement with a symmetric encryption scheme. They are also used in several integer factorization algorithms that have applications in cryptography, such as Lenstra elliptic-curve factorization.

Efficiency: *ECC* provides strong security with shorter key lengths compared to other public key cryptography systems, making it more efficient for mobile devices and resource-constrained environments.

Resistance to Quantum Attacks: *ECC* is considered more resistant to quantum attacks compared to some traditional public key algorithms.

Curve Diffie-Hellman (ECDH) is a variant of the *Diffie–Hellman* protocol using elliptic-curve cryptography. It is a key agreement protocol that allows two parties, each having an elliptic-curve public–private key pair, to establish a shared secret over an insecure channel. This shared secret may be directly used as a key, or to derive another key. The key, or the derived key, can then be used to encrypt subsequent communications using a symmetric-key cipher. [12] [13]

4. Which modes are used

- *Advanced Encryption Standard (AES):*

Signal uses *AES* for symmetric encryption, which is a widely adopted and secure symmetric encryption algorithm. Depending on the specific implementation, different modes of *AES* may be used, such as *Cipher Block Chaining (CBC)* or *Galois/Counter Mode (GCM)*.

5. How group chats are managed

The first time a WhatsApp group member sends a message to a group:

1. The sender generates a random 32-byte Chain Key.
2. The sender generates a random 32-byte Chain Key.
3. The sender combines the 32- byte Chain Key and the public key from the Signature Key into a Sender Key message

Combining means it just combines chain key and Signature public key to create Sender key but later receivers can decompose this Sender key into chain key and Signature public key.

4. The sender individually encrypts the Sender Key to each member of the group.

For all subsequent messages to the group:

1. The sender derives a Message Key from the Chain Key, and updates the Chain Key.
2. The sender encrypts the message using AES256 in CBC mode.
3. The sender signs the ciphertext using the Signature Key.
4. The sender transmits the single ciphertext message to the server, which does server-side fan-out to all group participants.

The “hash ratchet” of the message sender’s Chain Key provides forward secrecy. Whenever a group member leaves, the session is shut down and starts over again in the Setup Phase. Group participants clear their Sender Key and start over. This is done because former member has learned everybody's Sender keys so they clear their old sender key and exchange new sender key with each other. [6]

6. Role of symmetric and asymmetric encryption

Symmetric cryptography and asymmetric cryptography are two different types of encryption techniques used to secure data and communications.

- Symmetric cryptography, also known as secret key cryptography, uses a single secret key for both encryption and decryption. This means that the same key is used to encrypt and decrypt the data, and the key must be exchanged between the sender and the receiver before any communication can take place. This method is fast and efficient, but the key must be kept secret to maintain security.
- Asymmetric cryptography, also known as public key cryptography, uses a pair of keys: a public key and a private key. The public key is used to encrypt the data, and the private key is used to decrypt the data. The public key can be freely shared with anyone, while the private key must be kept secret. This method is more secure, but it is also slower and more resource-intensive than symmetric cryptography.

When deciding which type of encryption to use, it is important to consider the specific requirements of your application. In general, symmetric cryptography is best for encrypting large amounts of data or for situations where performance is a concern, while asymmetric cryptography is best for situations where security is a higher priority and where the exchange of keys is required. [14]

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